

===== DECISION TREE MASTER SHEET =====

1 Probability

$P = \text{Favorable Cases} / \text{Total Cases}$

2 Conditional Probability

$P(A|B)$

Example

Purchase Given Male

Purchase Given Age>30

3 Entropy

$\text{Entropy} = -\sum (P_i \times \log_2 P_i)$

Binary Formula

Entropy

$= -(P_{\text{yes}} \times \log_2 P_{\text{yes}})$

$$-(P_{no} \times \log_2 P_{no})$$

Pure Node

Entropy = 0

50-50

Entropy = 1

Entropy ↓ Better

4 Gini Index

$$\text{Gini} = 1 - (P_1^2 + P_2^2)$$

Pure Node

Gini = 0

50-50

Gini = 0.5

Gini ↓ Better

5 Information Gain

IG

=

Parent Entropy

-

Weighted Child Entropy

IG ↑ Better Split

6 Confusion Matrix

Predicted

Yes No

Actual Yes TP FN

Actual No FP TN

7 Accuracy

(TP+TN)

/

(TP+TN+FP+FN)

8 Precision

TP

/

TP+FP

9 Recall

TP

/

TP+FN

10 Specificity

TN

/

TN+FP

1 1 F1 Score

2PR

/

P+R

1 2 Error Rate

FP+FN

/

Total

1 3 ROC

Graph

$X = \text{FPR}$

$Y = \text{TPR}$

1 4 AUC

Area Under ROC

1 = Perfect

0.9 = Excellent

0.8 = Very Good

0.7 = Good

1 5 Train Accuracy

Training Data

1 6 Test Accuracy

Testing Data

1 7 Overfitting

Train High

Test Low

1 8 Underfitting

Train Low

Test Low

1 9 Good Model

Train $\approx$ Test

Difference kam

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